MIL-C-48811(PA)
9 June 1975
SUPERSEDING
See Section 6

MILITARY SPECIFICATION
CONNECTOR PLUG, ELECTRICAL AND CONNECTOR RECEPTACLE, ELECTRICAL

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for multi-connection type electrical connectors, plugs, and receptacles which are used in the Nike Hercules warhead section.

1.2 Classification. This specification covers Connectors, plugs and receptacles of the following.

Class A - solid shell
Class B - split shell
Class E - environment resisting

Types:

Plugs straight - conforming to drawings (dwg):

8791618 (Class A and B)
8791454 (Class A and B less coupling ring)
8795956 (Class E)
8788486 (Class E less coupling ring)

Angle 90 degrees - conforming to dwg:

8791803 (Class A and B)

Receptacles cable connecting - conforming to dwgs:

8786126 (Class A and B)
8791455 (Class A and B, steel shell, less coupling thread)
8785485 (Class E)

Receptacles box mounting - conforming to dwg:

8791453 (Class A and B)

Sizes:

As shown on shell drawings: 8791618, 8791454, 8795956, 8785486, 8791803, 8786126, 8791455, 8785485 and 8791453.

Arrangements:

As shown on insert drawings: 8791787 through 8791800 inclusive, 8804013 through 8804017 inclusive, 8804143, 8805872, and 8801698.

Styles:

Style P - Inserts containing pin contacts
Style S - Inserts containing socket contacts

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-P-416 - Plating, Cadmium (Electrodeposited).
QQ-A-591 - Aluminum-Base-Alloy; Die Castings.

MILITARY

MIL-M-14 - Plastic Materials, Molding, and Plastic Parts Molded; Thermosetting.
MIL-R-3065 - Rubber and Synthetic Rubber Compounds, General Purpose.
MIL-G-3278 - Grease, Aircraft and Instruments.
MIL-W-5086 - Wire, Electrical, 600-Volt, Copper, Aircraft.
MIL-W-5088 - Wiring, Aircraft, Installation Of.
MIL-G-5572 - Fuel, Aircraft, 100/130.
MIL-E-5624 - Fuel, Jet Aircraft, Grade JP-4.
MIL-O-5606 - Oil; Hydraulic, Aircraft, Petroleum Base.
MIL-L-6082 - Oil, Engine, Aircraft, Grade 1000.
MIL-F-7083 - Fluid, Hydraulic, Aviation.
MIL-S-7742 - Screw Threads; Standard Aeronautical.
MIL-L-7808 - Lubricating Oil, Gas Turbine, Aircraft.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Components Parts.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1168 - Lot Numbering of Ammunition.
MIL-STD-1169 - Packaging, Packing and Marking of Inert Ammunition Components.
MIL-STD-1460 - Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

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DRAWINGS

ARMY BALLASTIC MISSILE AGENCY

10962319 - Receptacle J-1
10962320 - Receptacle J-2
10962321 - Receptacle J-3
10962322 - Receptacle J-4
10962391 - Connector
10962392 - Connector
10962402 - Connector
10962403 - Connector
10962404 - Connector

ORDNANCE CORPS

TA-25-223 - Connector.
400539 - Connector Receptacle.
8785485 - Specification for Connector, Plug, Electrical.
8785486 - Specification for Connector, Plug, Electrical.
8786126 - Specification for Connector, Receptacle, Electrical.
8791618 - Specification for Connector, Plug, Electrical.
8791453 - Specification for Connector, Receptacle, Electrical.
8791454 - Specification for Connector, Plug, Electrical.
8791455 - Specification for Connector, Receptacle, Electrical.
8791803 - Specification for Connector, Plug, Electrical.
8795956 - Specification for Connector, Plug, Electrical.
8791787 - Insert 15 P or S for 20 Shell.
8791788 - Insert 1 P or S for 18 Shell.
8791789 - Insert 10 P or S for 16 Shell.
8791790 - Insert 14 P or S for 22 Shell.
8791791 - Insert 27 P or S for 20 Shell.
8791792 - Insert 23 P or S for 20 Shell.
8791793 - Insert 7 P or S for 20 Shell.
8791794 - Insert 15 P or S for 28 Shell.
8791795 - Insert 12 P or S for 28 Shell.
8791796 - Insert 28 P or S for 24 Shell.
8791797 - Insert 13 P or S for 18 Shell.
8791798 - Insert 21 P or S for 28 Shell.
8791799 - Insert 1 P or S for 16 or 16S Shell.
8791800 - Insert 7 P or S for 24 Shell.

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8800702 - Connector.
8800824 - Receptacle Box Mounting.
8800851 - Connector.
8801698 - Insert 16 P or S for 20 Shell.
8804013 - Insert 19 P or S for 22 Shell.
8804014 - Insert 5 P or S for 24 Shell.
8804015 - Insert 11 P or S for 28 Shell.
8804016 - Insert 7 P or S for 32 Shell.
8804017 - Insert 5 P or S for 36 Shell.
8804143 - Insert 5 P or S for 14S Shell.
8805872 - Insert 16 P or S for 20 Shell.
8813411 - Connector.
8813431 - Connector.
8813462 - Plug.
8813483 - Receptacle Push-Pull.
8813486 - Connector.
8813491 - Connector.
8813492 - Connector.
8813493 - Connector.
8813494 - Receptacle Pressurized Bulkhead.
8813496 - Connector.
8813497 - Plug Connector.
8813498 - Connector.
8813499 - Connector Receptacle Electrical, Radar Fuze Test.
8813500 - Connector.
8813501 - Connector.
8813502 - Connector.
8813511 - Connector.
8813512 - Connector.
8813597 - Cable Assembly, XT4087-8.
8813611 - Plug Straight.
8821493 - Plate Guide.
8822459 - Connector.
8826389 - Terminal Board and Connector Assembly.

PUBLICATIONS

Handbook H28 - Screw Thread Standards for Federal Services.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

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3.2 Assembly. The connectors shall comply with all requirements specified on the drawing (dwg) (see list of dwgs in 2.1), with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Materials requirements. Materials which are not covered by applicable specifications, or which are not specifically described herein shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Shells and coupling rings. Shells and coupling rings, except shells conforming to drawing 8791455, shall be made from high grade aluminum alloys. Die castings and workmanship shall conform to QQ-A-591, composition No. 1,2, 5,5A, 10 or 11. Shells conforming to drawing 8791455 shall be made from steel and shall be such as to enable the connector to meet the applicable requirements of this specification.

3.4.2 Insert material.

3.4.2.1 Rigid insert material. Rigid or plastic inserts shall be of a high grade dielectric, and shall be such as to enable the connector to meet the requirements of this purchase description. Materials conforming to MIL-M-14, Types CMG or MDG, are acceptable.

3.4.2.2 Resilient insert material. Resilient or rubber inserts shall be made from polychloroprene compound conforming to MIL-R-3065, Class SC717-A, B and E except that the durometer hardness shall be 72 to 85 Shore A.

3.4.3 Contacts. Contacts shall be made from suitable conductive materials. All contacting surfaces shall be smooth and shall be silver or gold over silver plated. Current ratings shall be as specified in Table I.

Table I
Current ratings

Contact Size	Amperes*
16	22
12	41
8	73
4	135
0	245

*Currents are based on single conductor in free air. For current carrying capacity of conductors in cables or bundles refer to MIL-W-5088.

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3.4.4 Tinning solder cups. Unless gold plated, the interior surface of solder cups for contact sizes 16 and 12 shall be tinned to at least 50 percent of their depth, and, for sizes 8,4, and 0, the interior shall be completely tinned with solder having a nominal content of 60 percent tin and 40 percent lead, or better grade of commercial solder. Only rosin, or alcohol and rosin shall be used as a flux, and all excessive flux shall be removed. No excess solder shall be on the exterior of the solder cups.

3.4.5 Finish. All metal parts other than electrical contacts shall be cadmium plated in accordance with QQ-P-416, Type II, Class 3, except that a preliminary plating of other metal is permissible, and that the corrosion requirements shall be in accordance with 4.4.3.9. The resulting finish shall be dull olive drab in color and shall be electrically conductive.

3.5 Design and construction. Connectors shall be designed and constructed to meet the requirements of this purchase description, and to conform to the drawings listed in 2.1. Detail parts of plugs and receptacles shall be securely attached within their shells to insure proper mating.

3.5.1 Contact design. Contact design shall be such that the pins and the sockets will not be damaged by any possible twisting or forcing during the process of mating connectors.

3.5.1.1 Pin engaging end. The entering end of pins, sizes 16, 12, and 8, shall be formed with a spherical radius approximately 1/2 the diameter of the pin, allowing for a flat not in excess of 0.015-inch diameter in the center of the spherical development. The tips of the spherical end of pins, sizes 4 and 0, shall be cut off to form a blunt end such that the diameter of the blunt end shall be 1/8 inch less than the pin diameter, as shown in Figure 1.

3.5.1.2 Socket engaging end. The entering end of the socket contact shall be rounded or chamfered to allow for directing and centering of the entering pin. The socket contact shall provide the spring action for maintaining the contacting pressure between the pin and socket. Size 12 and smaller socket contacts shall be designed to exclude the entrance of a pin 0.005 inch larger than the allowable maximum diameter of a mating pin.

3.5.1.3 Solder cups. The locations of the solder cups shall be such that the normal soldering operations shall not impair any of the parts of the assembly. The solder cup shall be as indicated in Figure 1. All solder cavities shall be so designed that liquid solder will not escape during normal soldering operations and interfere with the float of the contact. Socket type contacts shall be constructed so that liquid solder in the solder cup cannot leak through to the front and prevent insertion of the pin.

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3.5.1.4 Dimensions. Solder-type contact dimensions shall conform to Table II and Figure 1. The illustrations in Figure 1 are for dimensional purpose only and are not intended to indicate the design.

Table II
Solder-type contact controlled dimensions, in inches

Contact Size	A 1	B 2	C 3	D	E	F	
		+0	max	min	+0.063	min	max
	+0.001	-1/32			-0.000		
16	0.0625	1/2	0.127	0.069	0.250	0.096	0.116
12	0.094	11/16	0.190	0.112	0.375	0.130	0.150
8	0.142	11/16	0.300	0.205	0.500	0.243	0.259
4	0.225	11/16	0.441	0.328	0.625	0.370	0.397
0	0.357	11/16	0.597	0.464	0.625	0.510	0.550

Letters A,B,C,D,E, and F refer to Figure 1.

a. Dimension A is measured over plating.
b. Dimension B is meant to control the length of pin engagement. When resilient insert materials are used, B may be measured when insert is compressed.
c. These values are used for calculating mechanical spacing between contacts, and between contacts and shell.

3.5.1.5 Point of spring engagement. The distance from the front of the shell to a line representing the point at which a square ended pin of mean diameter starts to engage the spring member of the socket shall be no greater than the following values:

Contact Size	Distance From Shell to Point of Engagement
16	0.281 inch
12	0.375 inch
8	0.375 inch
4	0.375 inch
0	0.281 inch

3.5.2 Insert design. Inserts shall be designed and constructed with the proper sections and radii in order that they will not readily chip or break in assembly or in normal service. Hollow type inserts shall not be used. Depressions on front and rear pin insert disks and rear socket insert disks, used to achieve creepage distance, shall not be located directly opposite on front and rear surfaces of individual disks when resultant reduction in cross sectional area would cause structual weakness. In order to prevent cross plugging

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of alternate positions, socket insert front face contact entrance holes and chamfer, if used, shall be as small as practicable. Socket inserts shall provide suitable protection against a socket contacting a pin before the plug has been polarized with its receptacle.

3.5.2.1 Insert retention (not applicable to resilient inserts). On removable inserts, provision shall be made to hold the insert within the shell when the conduit nut or adapter is removed.

3.5.2.2 Insert positioning (not applicable to resilient inserts). Front and rear removable inserts shall be keyed to prevent rotation with respect to the shell, and over-all side play or the difference between insert keyway and shell key dimensions shall not be greater than 0.010 inch for sizes 12S through 16, or ± 1/2 degree from centerline for sizes 18 through 48.

3.5.2.2.1 Alternate insert positioning (not applicable to resilient inserts). Inserts may be provided with slots and markings for positioning in any of the angular positions indicated on the applicable Ordnance drawing. If provided, alternate position slots shall be approximately 1/16 inch shorter than the insert in order that the insert cannot be placed in an alternate position without first removing this obstruction. When the shoulder is 3/32-inch thick or less, these slot lengths shall be one half the thickness of the shoulder. Alternate position letters shall be permanently marked in clearly recognizable circles on the front face of size 18 inserts and larger, as shown in figure 2, unless specifically exempted by Ordnance drawings. Unless otherwise required by the procuring activity, connectors shall be furnished with the insert in normal position. Connectors furnished in an alternate position may have a full length normal positioning slot.

3.5.2.3 Contact arrangements. Contacts shall be arranged in accordance with the applicable drawing. Unless otherwise specified on the drawing, mechanical spacing and creepage distances shall not be less than as specified in Table III, except that for Class I, and resilient insert Class A or B the contact spacing tolerance shall be 0.002 inch.

Table III
Contact arrangement

Service Rating	Mechanical Spacing (Nominal) (Inches)	Creepage Distance Across The Insert Face (Nominal) (Inches)
INST		1/16
A	1/16	1/8
D	1/8	3/16
E	3/16	1/4
B	1/4	5/16
C	5/16	1

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3.5.2.4 Contact-alignment (not applicable to resilient insert types). To facilitate self alignment of mating contacts, inserts for socket contacts should be so designed that individual contacts will have over-all side play of 0.008 to 0.015.

3.5.3 Snap ring (not applicable to resilient insert types. Snap rings used to assemble the insert within the shell shall be readily accessible, suitable for repeated use, and shall not bend, twist, or pull out of position under normal service use. Snap rings shall be rectangular or square in cross section.

3.5.3.1 Snap ring slots (not applicable to resilient insert types). Snap ring slots shall have square sides and sufficient depth and width to prevent the snap ring from sliding, pulling or twisting out in service. One or more suitable recessed or extraction slots shall be provided on all sizes to insure easy removal of the snap ring from its slot.

3.5.4 Coupling connection. Coupling rings shall be so designed that the pin and socket contacts will engage or disengage as the ring is respectively tightened or loosened.

3.5.4.1 Safe-tying of coupling rings. All threaded coupling rings shall be designed for safety wiring in order to prevent the ring from loosening or vibrating free from its mating thread. At least one hold shall be provided for connector sizes 14 and smaller and at least three holes, equally spaced, for connector sizes 16 and larger of a diameter sufficient to accommodate a 0.032 wire.

3.5.5 Polarization. Polarization of the plug with its receptacle shall be accomplished by an integral key and shall be of such design that it cannot possibly interfere with the functioning of the coupling threads. The integral key shall be of sufficient size in order that it will be impossible to mate the connectors in more than one position. The mating groove shall be approximately rectangular in cross section. The polarization of the plug, with its receptacle, shall be accomplished before engaging the coupling rings.

3.5.6 Ease of mating. Plugs and receptacles of all arrangements shall be capable of being fully mated and unmated by hand, without the use of tools, at temperature extremes listed in MIL-STD-202, Method 102A, test condition A.

3.5.7 Screw threads. Screw threads intended to mate with parts of another approved manufacturer shall be UNEF or NEF Class 2A or 2B, conforming to MIL-S-7742, except that 1-3/4-18 and 2-18 threads shall be Class NS-2A or NS-2B, conforming to Handbook H28 and supplement.

3.5.7.1 Screw threads shall be checked after plating by means of ring and plug gages only in accordance with Handbook H28. Slight out-of-roundness beyond the tolerances of MIL-S-7742 is not objectionable if the threads can be checked without forcing of the thread gages.

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3.5.7.2 Screw threads may be relieved provided such relief does not interfere with proper performance of screw threads. When screw threads are relieved by under cutting or equivalent means, the relief dimension may be considered as a part of the useable thread dimension, provided that there be at least three full engagement threads.

3.5.7.3 All threads shall be coated with a lubricant conforming to MIL-G-3278.

3.6 Interchangeability (not applicable to resilient insert types). Removable insert assemblies of any one manufacturer shall be interchangeable between all shells of the same size and class made by that manufacturer.

3.7 High potential. Electrical connectors shall withstand the voltage indicated in Table IV applied between the two closest contacts and between the shell and the contacts closest to the shell for a period of one minute when tested as specified in 4.4.1.

Table IV
Voltage for high potential test

Service Rating	Test Voltage (rms) (60 Hz)
INST	1000
A	2000
D	2800
E	3500
B	4500
C	7000

3.8 Insulation resistance. The insulation resistance shall be 5000 megohms minimum between any pair of contacts and between the shell and any contacts when tested as specified in 4.4.2.

3.9 Environmental testing. The connectors shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The connectors shall also comply with 3.7 and 3.8 before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

3.9.1 Thermal shock. The plugs and receptacles shall withstand exposure to temperature changes for a total of five cycles performed continuously when tested as specified in 4.4.3.1. The insert shall not crack and the connector shall meet the high potential requirements of 3.7 within 45 minutes after completion of the final cycle.

3.9.2 Contact retention. The individual contacts shall withstand axial loads in either direction as shown in Table V, when tested as specified in 4.4.3.2.

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Table V
Contact retention

Contact Size	Axial Load (minimum)
16	10 lb
12	15 lb
8	20 lb
4	20 lb
0	25 lb

3.9.3 Insert retention (resilient inserts). The resilient inserts shall be capable of withstanding an axial load as specified in Table VI in either direction without being dislocated from their normal position in the connector shell when tested as specified in 4.4.3.3.

Table VI
Resilient insert retention test pressure

Connector Size	Gage Pressure (psi)
8 through 12	150
14 through 18	100
20 through 22	75
24 through 28	60
32 through 40	45

3.9.4 Vibration. A complete connector assembly shall be capable of withstanding the vibration test. The plug shall not become disengaged from the receptacle, there shall be no cracking, breaking, or loosening of parts when subjected to this test, and there shall be no interruption of continuity of any of the contact circuits when tested as specified in 4.4.3.4. After this test the plugs and receptacles shall meet the requirements of high potential as specified in 3.7.

3.9.5 Impact strength of rigid inserts (socket type). Connectors containing socket type plastic inserts shall be capable of withstanding, without cracking or chipping, the physcial impact of a 66.5 ± 2 gram, 1-inch diameter solid steel ball dropped upon the front face of the insert from a free height of 18 inches, when tested as specified in 4.4.3.5.

3.9.6 Humidity (not applicable to Class E). Mated plugs and receptacles shall withstand exposure to a temperature of 160 ± 3.6°F and a relative humidity of 95 ± 3 percent for a period of 14 days minimum when tested as specified in 4.4.3.6. Immediately after exposure, without any forced drying, the mated connectors shall meet a potential test of 1 1/2 times the ac voltage specified in Table VII for a period of 5 minutes.

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Table VII
Basic voltages, humidity and corrosion tests

Service Rating	AC Volts (rms)
INST	200
A	500
D	900
E	1250
B	1750
C	3000

3.9.7 Moisture resistance (Class E only). Mated plugs and receptacles shall withstand the moisture cycling test of MIL-STD-202, method GA, except that step 7a shall be at -65 + 3.6°F and step 7b shall be deleted, when tested as specified in 4.4.3.7. Engaged connectors after being tested and while still subjected to high relative humidity conditions shall comply with the insulation resistance requirements of 3.8 except that the insulation resistance shall be 100 megohms minimum.

3.9.8 Durability. The complete connector assembly, less coupling ring, shall be capable of withstanding 500 cycles minimum of insertions and withdrawals at a rate of 600 cycles per hour minimum, when tested as specified in 4.4.3.8. After 500 cycles, the plug and receptacle assembly shall show no electrical or mechanical defects, and the contacts shall meet the requirements of the resistance of contacts test as specified in 3.9.12.

3.9.9 Immersion. The connectors shall be capable of immersion in the following fluids for the indicated time-period:

(a) Aircraft fuel 100/130 conforming to MIL-G-5572: 20 hours minimum.
(b) Aircraft engine oil, Grade 1100 conforming to MIL-L-6082: 20 hours minimum.
(c) Aviation hydraulic fluid (petroleum base) conforming to MIL-O-5606: 20 hours minimum.
(d) Isopropyl alcohol: 20 hours minimum.
(e) Carbon tetrachloride: 2 hours minimum.
(f) Aviation hydraulic fluid conforming to MIL-F-7083: 20 hours minimum.
(g) Aircraft lubricating oil conforming to MIL-L-7808: 20 hours minimum.
(h) Jet aircraft fuel conforming to MIL-E-5624: Grade JP-4: 20 hours minimum.

After immersion the matability of the connectors shall not be affected and the assembled connectors shall withstand the voltage specified in Table VIII for five minutes minimum.

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Table VIII
Immersion test voltages

Service Rating	AC Volts (rms)
INST	400
A	1000
D	1400
E	1750
B	2250
C	3500

3.9.10 Cable twisting (Class E). The maximum allowable torque transmitted by the clamp to the cable shall not exceed that specified in Table IX, when tested as specified in 4.4.3.10.

Table IX
Cable twisting test (Class E)

Mating Shell Size	Test Dowel (in.) (Tol + 0.015)	Torque Applied to Clamp Nut (Inch Pounds min)	Maximum Allowable Torque Transmitted to Dowel (Inch Ounces)
8S,10S	0.157	40	1
10SL, 12	0.250	50	1
14	0.376	65	1
16	0.453	80	1
18	0.552	100	1
20, 22	0.676	150	2
24, 28	0.814	200	2
32	1.138	280	2
36	1.301	350	3
40	1.549	400	3

3.9.11 Key strength (applied to drawing 8791455 only). The polarization key and surrounding shell shall withstand the minimum torque specified in Table X applied along the entire side of the key without evidence of breaking or cracking.

Table X

Connector Size	Torque Minimum (inch pounds)
8S,10S,10SL,12,12S	130
14,14s,16,16S	200
18-20	250
22	300
24	350
28,32,36,40	400

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3.9.12 Resistance of contacts. The resistance at 77 + 5°F of pin and socket contacts when assembled as in service shall be such that the potential drop at the specified current shall not be greater than that indicated in Table XI, when tested as specified in 4.4.3.12.

Table XI
Contact potential drop

Contact Size	Test Current (Amperes)	Potential Drop (Millivolts)(Max)
16	20	21
12	35	20
8	60	12
4	110	10
0	200	10

3.9.13 Contact engaging and separating forces. Using a steel pin of minimum diameter, with 12 + 4 micro-inch finish, socket contacts in connectors under test shall meet the minimum and maximum engaging and separating force requirements of Table XII, when tested as specified in 4.4.3.13. Readings shall be taken with gradually increasing loads until the pin either engages within or separates from the socket contact.

Table XII
Contact engaging and separating forces

Contact Size	Force in Pounds (Maximum)	Force in Pounds (Minimum)
16	3	1/4
12	5	1/2
8	10	3/4
4	15	1
0	20	2

3.9.14 Resistance to arc. The resistance to arc of insert materials shall be a minimum of 115 seconds.

3.9.15 Dielectric strength. The dielectric strength shall be 100 volts per mil minimum, when tested as specified in 4.4.3.14.

3.9.16 Dimensional stability. Plastic insert materials, molded by the manufacturer without a post-mold treatment, shall be capable of maintaining a dimensional stability within 0.7 percent compared to original measurements after subjection to ten cycles of temperature conditions at 257 + 5°F 50 percent relative humidity for 24 hours, when tested as specified in 4.4.3.10.

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3.10 Marking. Each connector shall be clearly and permanently marked with the manufacturer's name or trademark.

3.10.1 Alternate insert position letter. The letter W,X,Y or Z, as applicable, shall be added to the part number after the style letter when an alternate position is specified by the procuring activity.

3.10.2 Marking of inserts. Inserts shall be marked as shown in Figure 2 and the applicable drawings.

3.10.2.1 Contact designation. Contacts shall be designated as shown on the applicable drawing. Letters shall be clearly arranged to avoid confusion between contacts, and shall be either raised or depressed. All letters shall appear on the front of each insert, and as many as practical on the rear face. Lettering of socket inserts shall correspond with that of the mating pin insert.

3.11 Soldering. All soldering shall comply with the requirements of MIL-STD-1460.

3.12 Workmanship.

3.12.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.12.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.12.3 Burrs. Parts shall be free from burrs.

3.12.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.12.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.12.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit the number of complete assemblies listed below (see dwg listed in 2.1) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.1.1 Test samples submitted shall consist of the following:

a. Two each completely assembled plugs or receptacles in each type, size, and class on which approval is desired.
   NOTE: Plug and receptacle assemblies are to be furnished complete with mating insert arrangements of the greatest complement of contacts within the applicable shell type, size, and class.
b. Five pin contacts and fifty-five socket contacts of each size on which approval is desired.
c. Six each insert material specimens 4" in diameter x 1/8" thick on which approval is desired.
d. Three each plastic insulator material specimen bars 5" x 1/2" x 1/2", on which approval is desired.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table XIII. Tests shall be conducted in order specified in Table XIII.

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Table XIII
First article inspection and tests

Inspection or Test	Dest	Non Dest	Samples Inspected (See Paragraph)
1. Classification of characteristics (see 4.3.2.1)		X	All
2. High potential (see 3.7)		X	4.2.1.1 a
3. Insulation resistance (see 3.8)		X	4.2.1.1 a
4. Thermal shock (see 3.9.1)	X		4.2.1.1 a
5. Contact retention (see 3.9.2)		X	4.2.1.1 a
6. Insert retention (*resilient inserts*) (see 3.9.3)		X	4.2.1.1 a
7. Vibration (see 3.9.4)	X		4.2.1.1 a
8. Impact strength (rigid inserts, socket style) (see 3.9.5)		X	4.2.1.1 a
9. Humidity (Class A & B) (see 3.9.6)	X		4.2.1.1 a
10. Moisture (Class E) (see 3.9.7)	X		4.2.1.1 a
11. Durability (see 3.9.8)	X		4.2.1.1 a
12. Corrosion (see 3.9.9)	X		4.2.1.1 a & b
13. Immersion (Class A & B) (see 3.9.10)	X		4.2.1.1 a
14. Cable twisting (Class E) (see 3.9.11)		X	4.2.1.1 a
15. Key strength (applicable to connectors dwg 8791455 only)(see 3.9.12)		X	4.2.1.1 a
16. Contact resistance (see 3.9.13)		X	4.2.1.1 b
17. Contact separation force (see 3.9.14)		X	4.2.1.1 b
18. Resistance to arc (see 3.9.15)		X	4.2.1.1 c
19. Dielectric strength (see 3.9.16)		X	4.2.1.1 c
20. Dimensional stability (see 3.9.17)		X	4.2.1.1 d

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

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4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table XIV and Table XV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table XIV
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table XV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Inserts (see applicable dwg 2.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Cracks or slits, open knit lines (rigid only) or nicks	Visual
102.	Surface and edge irregularities (rigid only)	Visual
103.	Cracks and blisters (resilient only)	Visual
104.	Voids	Visual
105.	Inclusion of conductive material	Visual
106.	Cracks, slits, or checking (resilient only)	Visual
107.	Conductive material visibly bridging contact to edge or contact to shell	Visual
Minor:		
201.	Cracks (resilient only) less than 1/4 of the distance	Visual
202.	Cond. mat'l of greater dimension equal to or greater than minimum contact spacing	Visual
203.	Cond. mat'l of greater dimension 1/4 of minimum contact spacing	Visual
204.	Workmanship (see 3.12)	Visual

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4.3.2.1.2 Die cast or machined parts (see applicable dwg 2.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Cracks or voids which pass at least half way through shells, cable clamps, fractures due to rolling or spinning (except feathered edges)	Visual
102.	Cracks or voids complete through backshell	Visual
103.	Objective evidence relative to material improper or missing	Visual
Minor:		
201.	Workmanship (see 3.12)	Visual

4.3.2.1.3 Pin contacts (see applicable dwg 2.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Burred, bent, misaligned or damaged to prevent mating	Visual/Manual
102.	Short or long pins (incorrect pins)	Gage/Manual
103.	Non-adherence of solder or no turning on inside of solder pots except gold flashed contacts	Visual
Minor:	None defined.	

4.3.2.1.4 Socket contacts (see applicable dwg 2.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Burred, bent, misaligned or damaged to prevent mating	Visual/Manual
102.	Non-adherence of solder or no tinning on inside of solder pots except gold flashed contacts	Visual

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Minor: None defined.

4.3.2.1.5 Plating (see applicable dwg 2.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Plating completely missing on sliding or mating surfaces of contacts	Visual
Minor:		
	201. Missing plating on other surfaces	Visual

4.3.2.1.6 Complete assemblies (see applicable dwg 2.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Illegible, incorrect or missing identification	Visual
	102. Incomplete or incorrect parts affecting interchangeability or function	Visual
	*103. Mislocation of pins, sockets, keys or keyways which cause damage to plug plus or receptacle sockets when mated	Gage/Visual
	104. Objective evidence relative to material improper or missing (see 3.1)	Visual

*Optical comparators, charts and templates shall be used to verify proper locations of pins.

Minor:		
	201. Workmanship (see 3.12)	Visual/Manual

4.3.2.1.7 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Preservation, packing and packaging in accordance with MIL-STD-1169.....Visual
102. Marking in accordance with MIL-STD-1169.....Visual

Minor:
201. Workmanship (see 3.12).....Visual/Manual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

	Test	Test Classification
a.	High potential	Major
b.	Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 High potential, insulation resistance. A sample shall be selected at random in accordance with Table XVI. The sample shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.7.1 and 3.7.2, the lot shall be rejected.

4.3.4 Acceptance inspection equipment. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions and calibration instructions prepared by the contractor. The contractor shall obtain approval of such designs, operating instructions and special calibration instructions as specified in 6.3 prior to use on the contract.

4.3.4.1 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 High potential. Electric connectors shall show no evidence of breakdown when the voltage indicated in Table XVII is applied between the two closest contacts, and between the shell and the contacts closest to the shell, for a period of one minute. The test voltage shall be applied gradually at the rate of approximately 500 volts each second until the

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specified voltage is reached. Requirements shall be as specified in 3.7.

Table XVII
Voltage for high potential test

Service Rating	Test Voltage (rms)(60 Hz)
INST	1000
A	2000
D	2800
E	3500
B	4500
C	7000

4.4.2 Insulation resistance. The insulation shall be greater than 5000 megohms, when measured separately between any pair of contacts, and between the shell and any contact. Measurements shall be made by an approved method using a potential of 500 volts. Electrification time shall not exceed two minutes. Tests shall be conducted at room temperature of 77 + 5°F. Requirements shall be as specified in 3.8.

4.4.3 Environmental testing (see 3.9).

4.4.3.1 Thermal shock. Plugs and receptacles shall be exposed to the temperature cycling test described in MIL-STD-202, Method 102A, test condition D. The insert shall not crack, and the connector shall meet the high potential requirements of 4.4.1 within 45 minutes after completion of the final cycle.

4.4.3.2 Contact retention. After submission to the thermal shock test as specified in 4.4.3.1 individual contacts shall withstand axial loads in either direction as shown in Table XVIII. Measurements shall be made on individual contacts with all other contacts in place and the insert in a shell. The load shall be applied uniformly at a rate of 1 pound per second. Requirements shall be as specified in 3.9.2.

Table XVIII
Contact retention

Contact Size	Axial Load (minimum)
16	10 lbs
12	15 lbs
8	20 lbs
4	20 lbs
0	25 lbs

4.4.3.3 Insert retention (resilient inserts). Resilient inserts shall withstand an axial load in either direction as specified in Table XIX without being dislocated from their

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normal position in the connector shell. Loading shall be accomplished by applying air pressure to either face of the insert. The pressure shall be increased gradually at a rate of approximately 10 pounds per second, until the specified pressure is reached. The insert assembly shall retain its normal location in the connector shell for 5 seconds at the specified pressure. Prior to this test, the plug or receptacle assembly shall be conditioned by subjecting it to the thermal shock test as specified in 4.4.3.1. For test purposes, the insert arrangement having the largest number of contacts of each size made by the manufacturer shall be used. Requirements shall be as specified in 3.9.3.

Table XIX
Resilient insert retention test pressure

Connector Size	Gage Pressure (psi)
8 through 12	150
14 through 18	100
20 through 22	75
24 through 28	60
32 through 40	45

4.4.3.4 Vibration. A complete connector assembly shall be mounted as shown in Figure 3 and vibrated in accordance with MIL-STD-202, Method 204, Test Condition A, vibration test. All contacts shall be wired with the proper wire size in accordance with MIL-W-5086. One pair of each size contacts, representative of the contacts used, shall be connected to a suitable testing circuit with at least 100 milliamperes flowing through the contacts. The plug and receptacle shall be held together only by the locking means normally provided. A test setup similar to Figure 3 shall be used. The plug shall not become disengaged from the receptacle, there shall be no cracking, breaking, or loosening of parts when subjected to this test, and there shall be no interruption of continuity of any of the contact circuits. After this test, the plugs and receptacles shall meet the physical requirements specified herein, 3.9.4 and 4.4.1. The inserts shall be inspected through a device having a magnification power of approximately three diameters, aided by any chemical agent to show cracks.

4.4.3.5 Impact strength of rigid inserts (socket type). Connectors containing socket type plastic inserts shall be capable of withstanding, without cracking or chipping, the physical impact of a 66.5 ± 0.2 gram, 1-inch diameter, solid steel ball dropped upon the front face of the insert from a height of 18 inches. Requirements shall be as specified in 3.9.5.

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4.4.3.6 Humidity (not applicable to Class E). Mated plugs and receptacles shall be exposed to a temperature of 160 + 3.6°F and a relative humidity of 95 + 3 percent for a period of 14 days. Immediately after exposure, without any forced drying, the mated connectors shall meet a potential test of 1-1/2 times the ac voltage specified in Table XX for a period of 5 minutes. Requirements shall be as specified in 3.9.6.

Table XX
Basic voltages, humidity and corrosion tests

Service Rating	AC Volts (rms)
INST	200
A	500
D	900
E	1250
B	1750
C	3000

4.4.3.7 Moisture resistance (Class E only). Mated plugs and receptacles wired with the proper wire size in accordance with MIL-W-5086, shall be tested in accordance with Moisture Resistance test described in MIL-STD-202, Method 106A, except that step 7a shall be at -67 + 3.6°F and step 7b shall be deleted. Engaged connectors after being tested and while still subjected to high relative humidity conditions shall comply with the insulation resistance requirements of 4.4.2 except that the insulation resistance shall be not less than 100 megohms. Requirements shall be as specified in 3.9.7.

4.4.3.8 Durability. Two complete connector assemblies, less coupling ring, shall be subjected to 500 cycles of insertion and withdrawal at a rate not exceeding 600 cycles per hour. The insertions and withdrawals shall be accomplished in a manner similar to that which the connectors shall be subjected to in service. After 500 cycles, the plug and receptacle assemblies shall show no electrical or mechanical defects. Requirements shall be as specified in 3.9.8.

4.4.3.9 Immersion. Two unmated connectors shall be immersed in the following fluids, one connector in each fluid, for the indicated time period:

a. Aviation hydraulic fluid (petroleum base) conforming to MIL-O-5606: 20 hours.
b. Aircraft lubricating oil, conforming to MIL-L-7808: 20 hours.

Upon removal from the fluid, each connector shall remain for a one-hour period in free air at room temperature. The connector specimens shall then be engaged with their mating parts,

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and the assembled connectors shall withstand the voltage specified in Table XXI for a period of five minutes.

Table XXI
Immersion test voltages

Service Rating	AC Volts (rms)
INST	400
A	1000
D	1400
E	1750
B	2250
C	3500

4.4.3.10 Cable twisting (Class E). The cable clamp shall be assembled to a standard back shell, or equivalent, with the cable replaced by a wooden test dowel to which is coupled a torque measuring device. With the specified test dowel and applied torque, the maximum allowable torque transmitted by the clamp to the cable shall not exceed that specified in Table XXII.

Table XXII
Cable twisting test

Mating Shell Size	Test Dowel Dia (In.) (Tol + 0.015)	Torque Applied to Clamp Nut (In.-Lbs) (Min)	Maximum Allowable Torque Transmitted to Dowel (In.-Ozs)
8S,10S	0.157	40	1
10SL,12	0.250	50	1
14	0.376	65	1
16	0.453	80	1
18	0.552	100	1
20,22	0.676	150	2
24,28	0.814	200	2
32	1.138	280	2
36	1.301	350	3
40	1.549	400	3

4.4.3.11 Key strength (applied to drawing 8791455 only). The polarization key and surrounding shell shall withstand the minimum torque specified in Table XXIII applied along the entire side of the key without evidence of breaking or cracking.

Table XXIII

Connector Size	Torque Minimum (Inch-Lbs.)
8S,10S,10SL,12,12S	130
14,14S,16,16S	200
18,20	250
22	300

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24	350
28,32,36,40	400

4.4.3.12 Resistance of contacts. The resistance at 77 + 5°F of pin and socket contacts when assembled as in service shall be such that the potential drop at the specified test current shall be not greater than that indicated in Table XXIV. The electrical resistance of each pair of mated pin and socket contacts in the sample connector assembly shall be determined by measuring the potential drop across the assembled contacts when carrying the specified test current. In making this test, the potential drop shall be measured at the extreme terminal end of the contact as shown in Figure 4.

Table XXIV
Contact potential drop

Contact Size	Test Current (Amperes)	Potential Drop After Corrosion (Millivolts) (Max)
16	20	35
12	35	30
8	60	25
4	110	20
0	200	20

4.4.3.13 Contact separation forces. Provisions shall be made for mounting socket contacts in a suitable position for applying gradually increasing loads during withdrawal of the minimum diameter hardened steel test pin. The finish on the test pin shall not exceed 10 micro-inches and the engaging end of the test pin shall be rounded or chamfered. The average force required to separate the test pin from the sockets shall not exceed the maximum average or be less than the minimum average values shown in Table XXV, and none of the values shall be less than the minimum force indicated in Table XXV. The values shall be obtained from a sample lot of 50 socket contacts. The separation force may be determined with sockets either in or out of their insulators. Gradually increasing loads shall be applied until the test pin properly separates from the socket. Depth of engagement shall conform to that encountered in service.

Table XXV
Contact engaging and separating forces

Contact Size	Test Pin Diameter + 0.0002 - 0.0000	Force in Lbs. Min Average	Max Avg	Min Ind
16	0.0615	0.234	0.150	0.125

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12	0.0930	0.625	2.75	0.375
8	0.1410	1.000	6.00	0.625
4	0.2240	1.250	9.00	0.750
0	0.3560	2.500	15.00	1.500

4.4.3.13.1 Resistance to test prod damage. Size 16 socket contacts shall meet the requirements of 4.4.3.13 after having been subjected to the following test: A test prod of hardened steel having a diameter equal to a nominal mating pin shall be inserted into each socket to (a) 1/2-inch (b) 3/8-inch, and (c) 1/4-inch depth. At each of these depths, measured from the face of the insert, a bending moment of 2 in. lb ± 10 percent shall be applied to the prod, about the inserted end of the prod, and the connector shall be rotated in one direction through 360 degrees, in order that a uniform force is applied to the inside surface of the socket. This test shall be performed with the socket contacts in their inserts, and the sockets locked, if necessary, to prevent rotation in the inserts during the test.

4.4.3.13.1.1 Test procedure for resistance to test prod damage. In order to insure uniform test results, the test fixture shown in Figure 5 shall be used to perform the test required by 4.4.3.13.1. Procedures shall be as follows:

(a) With the weight in position for the 2 inch-pound moment and no spacers on the pin, insert the pin in a socket of an assembled receptacle or plug while the axis of the socket is in a horizontal position.
(b) With the fixture free and unsupported, rotate the receptacle or plug 360° about a horizontal axis maintaining the socket in a horizontal position.
(c) Repeat with size 1 spacer.
(d) Repeat with size 2 spacer.
(e) After withdrawal of the fixture, the socket shall pass the engaging and separating loads of 4.4.3.13.

4.4.3.14 Resistance to arc. The resistance to arc of insert material disk specimens conforming to designation 2 of MIL-M-14 shall be a minimum of 115 seconds when tested in accordance with MIL-M-14; except as may otherwise be specified below, using the circuit illustrated, and the following conditions:

Conditions:

(a) 12 kv (minimum) transformer.
(b) kva rating of transformer not specified, but 10 ma (rms) must be obtainable.
(c) Sufficient external resistance may be added in the leg of the circuit that is not grounded, if necessary, to obtain the required 10 ma across the electrodes in air (not reading on test specimen).
(d) Cable on resistance side shall be supported in air.
(e) Floor or chamber shall have a 1-inch thick insulating sheet for supporting the specimen under test.

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(f) A milliammeter shall be placed in the circuit between the ground and the arc electrodes.
(g) Specimen shall be cleaned with a clean cloth dampened with alcohol, and its flat surface should not be handled thereafter until after the arc test.
(h) Arc electrodes shall be cleaned with a soft, clean cloth, dampened with alcohol, and dried with a soft, clean, dry cloth before each test.
(i) Primary voltage shall be controlled with a variable auto transformer (variac).
(j) Relative humidity shall be between 30 and 40 percent.
(k) Interrupter must cause an arc to flow 1/4 second and cease 1-3/4 seconds repetitively during the first minute and to flow 1/4 second and cease 3/4 second repetitively during the second minute.
(l) The electrodes shall consist of two steel rods 0.186-inch diameter, equipped with tungsten wire tips 0.060 inch in diameter, which have conical points with a 60 degree included angle. The electrodes shall be mounted to an insulated block at 45 degrees to the vertical, and shall be adjusted to give a gap of 0.320 inch. Both electrodes shall be in the same horizontal and vertical planes.

4.4.3.15 Dielectric strength. Insert material disk specimens conforming to Designation 2 of MIL-M-14 shall be conditioned in accordance with procedure E-48/50 + D-48/50 of that specification and subjected to the step-by-step test. Dielectric strength so determined shall be a minimum of 100 volts per mil.

4.4.3.16 Dimensional stability. Plastic insert material bars 5 by 1/2 by 1/2 inches, molded by the manufacturer without a post-mold treatment and so machined that the 1/2 by 1/2 inch ends are smooth and parallel, shall be submitted to test within a maximum of 10 days after removal from the mold. Conditioning shall consist of 96 hours at 77°F and 50 percent relative humidity. The initial length of the specimen shall then be cycled in a circulating air oven at 257 ± 9°F for a period of 48 hours followed by 77 ± 9°F and 50 percent relative humidity for 24 hours. After 10 complete cycles, the length of the specimen shall again be measured to the nearest 0.001 inch. The percentage change in length calculated as follows shall not exceed 0.7 percent:

$$\text{Dimensional change (percent)} \pm \frac{\text{initial length} - \text{final length}}{\text{initial length}} \times 100$$

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The connector shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The connector shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The connector shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

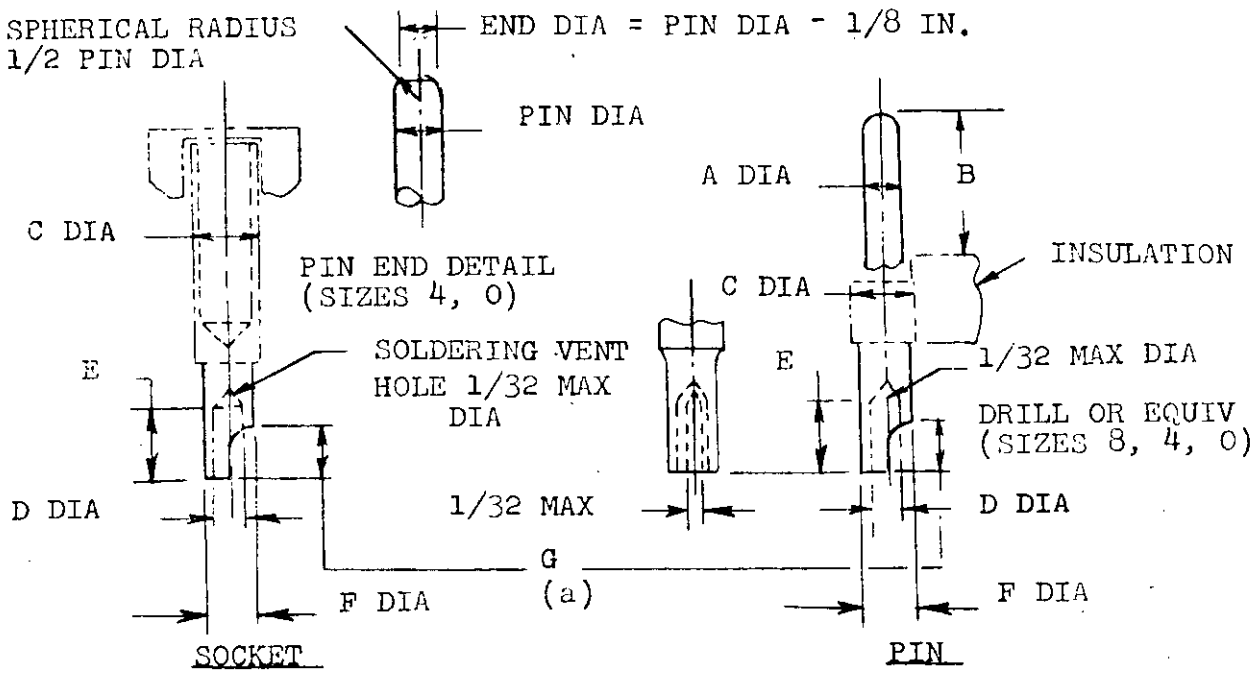
6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1014 dated 8 October 1959 and Amendment 1, dated 17 May 1960.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. NUOR-A039

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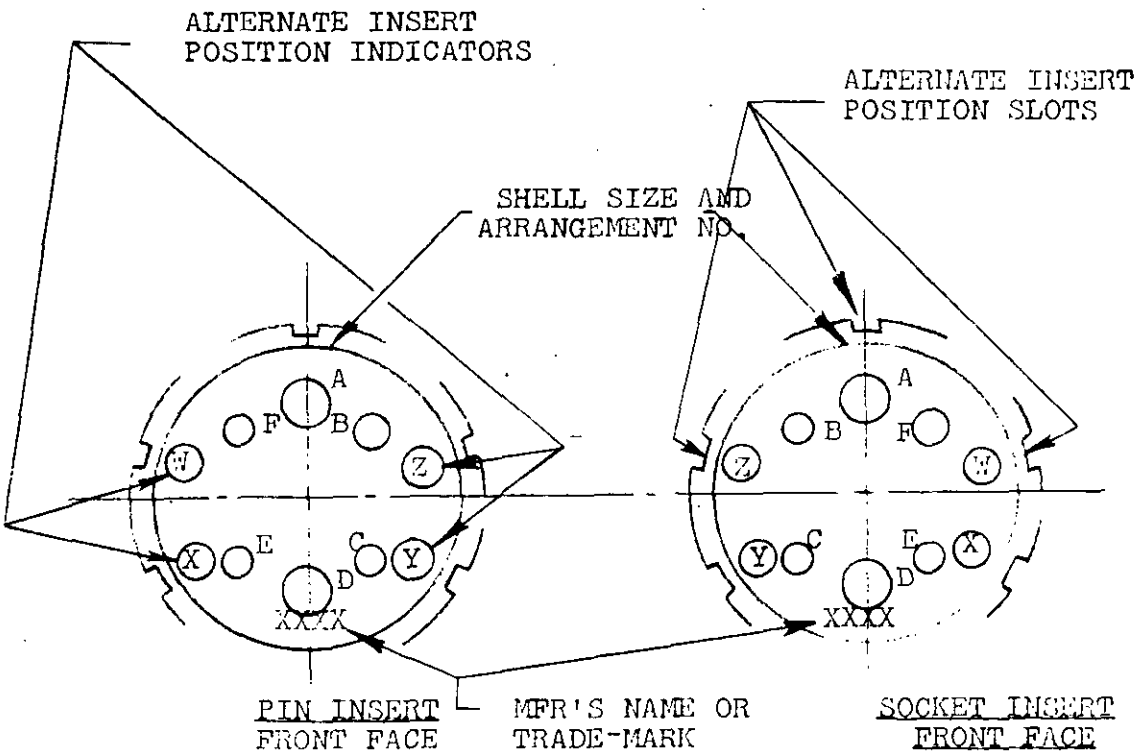


(a) "G" DIMENSION LIMITED TO A MAXIMUM OF 2/3 OF THE "E" DIMENSION.

DIMENSIONS IN INCHES

FIGURE 1

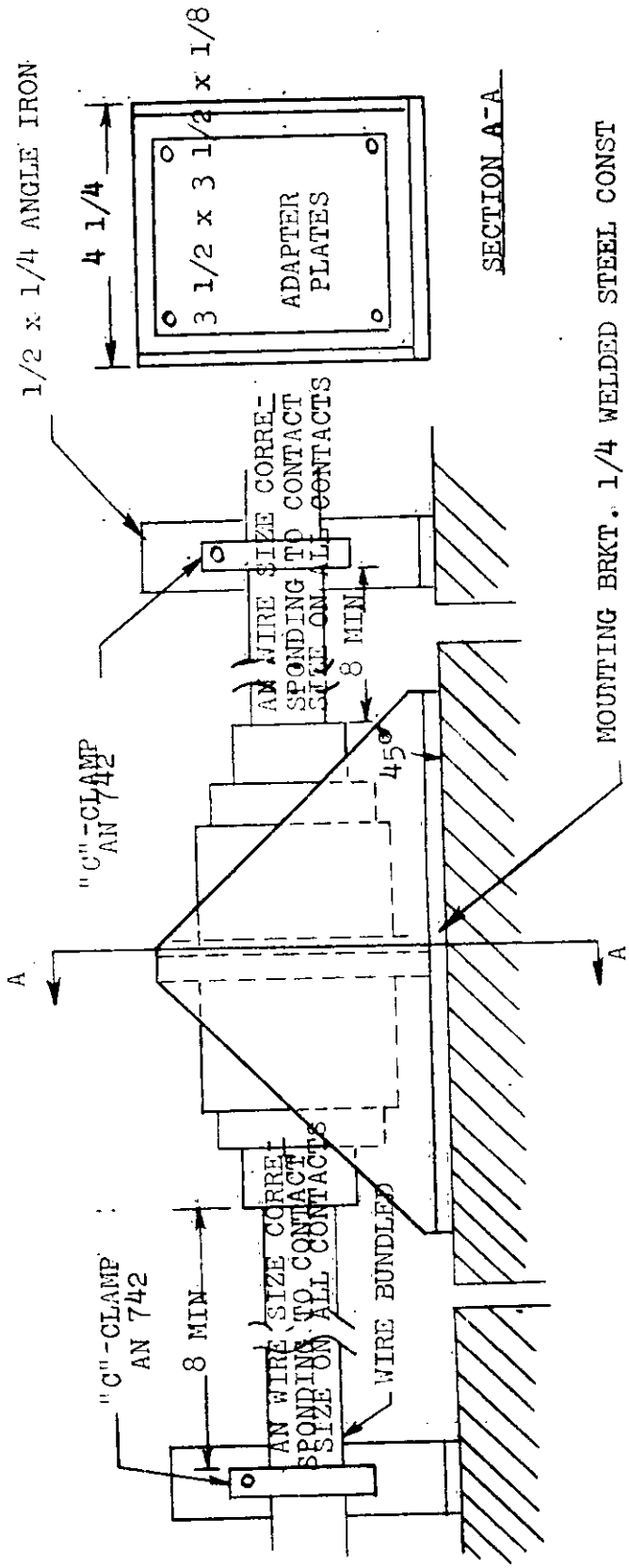
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VARIATION IN LOCATION OF SHELL SIZE, ARRANGEMENT NO., AND MFR'S NAME OR TRADE-MARK IS PERMITTED.

FIGURE 2

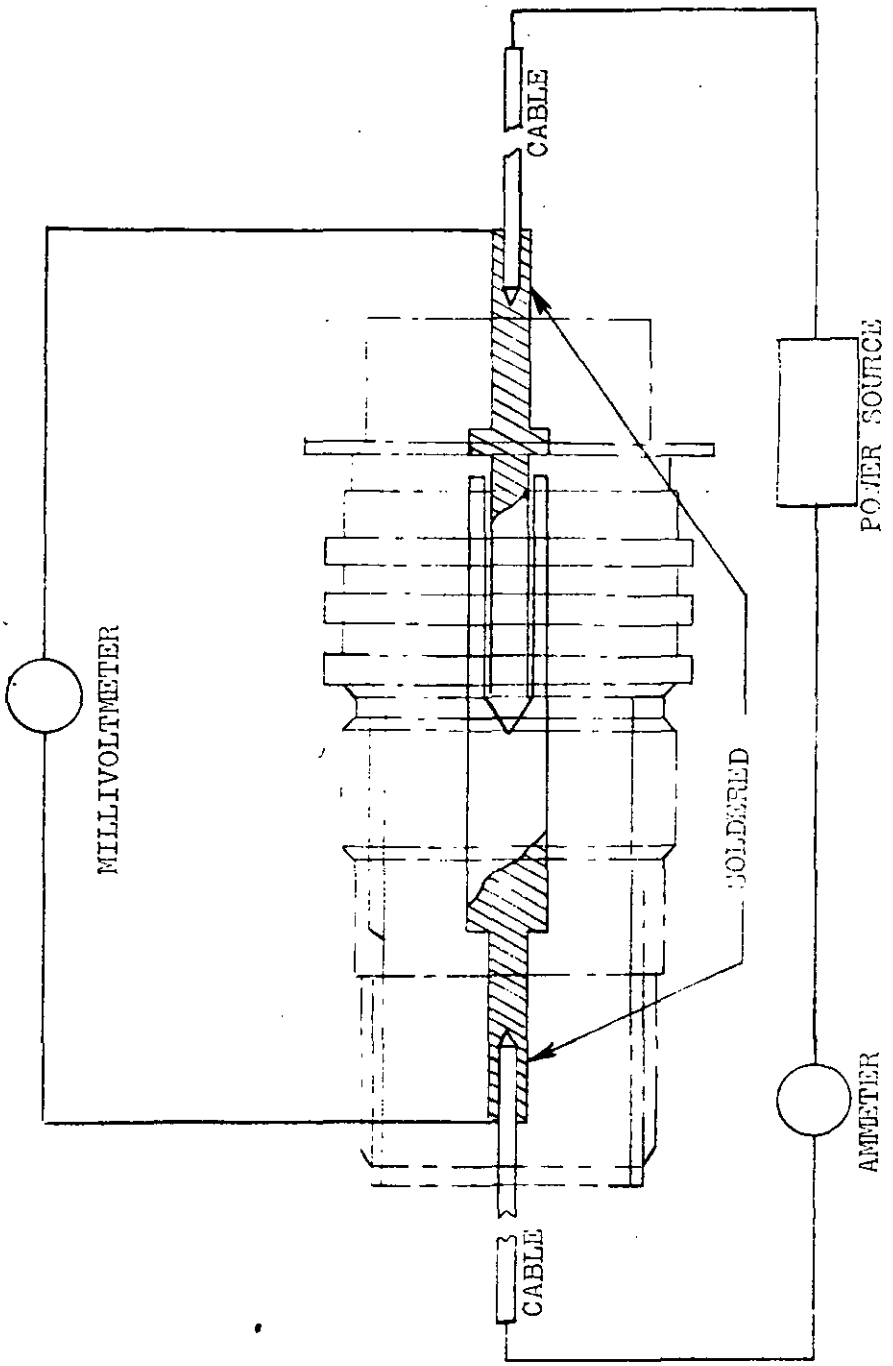
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VIBRATION TESTING SETUP

FIGURE 3

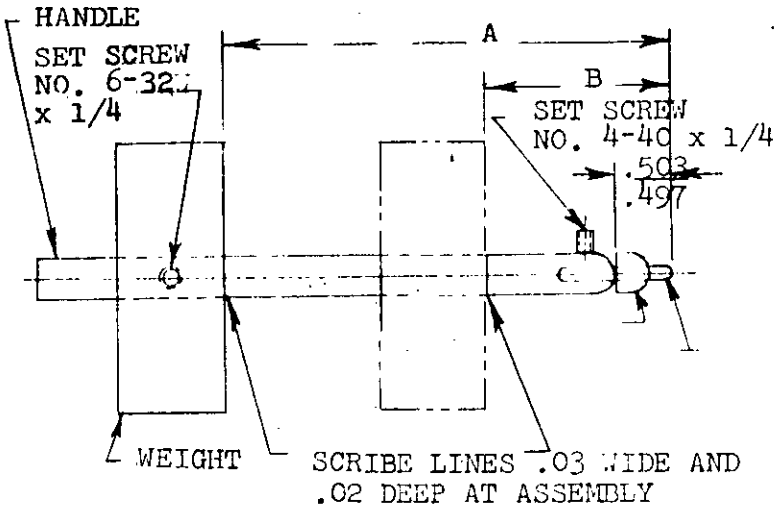
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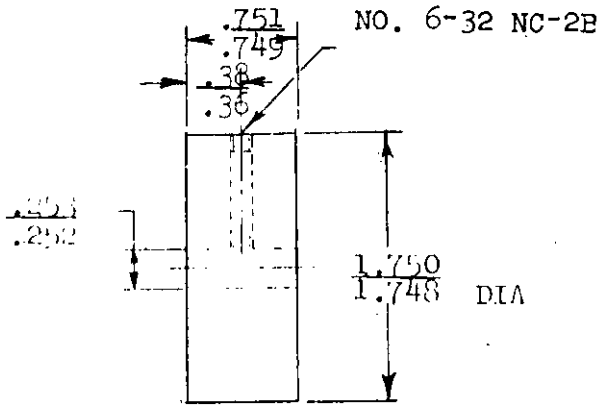
WIRING DIAGRAM FOR
RESISTANCE OF CONTACTS TEST

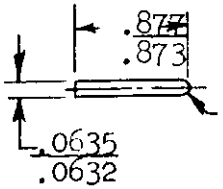
FIGURE 4

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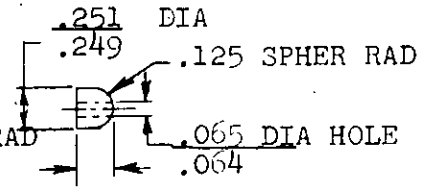
A = 3.05 (APPROX FOR 2 IN. LB)
B = 1.20 (APPROX FOR 1 IN. LB)



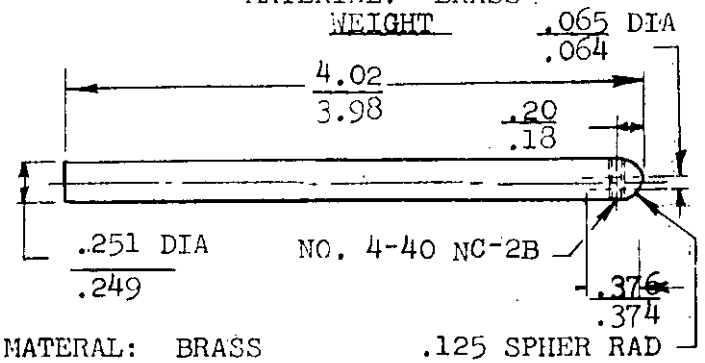


MATERIAL: HARDENED STEEL

PIN



MATERIAL: BRASS



MATERAL: BRASS

HANDLE

SIZE 1, A = .251/.248

SIZE 2, A = .126/.124

SPACERS

TEST FIXTURE FOR RESISTANCE TO TEST PROD DAMAGE TEST

FIGURE 5

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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